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Activity 3.1

Covalent radii (in Å) for some elements of different groups and periods are listed below. Plot these values against atomic number. From the plot, explain the variation along a period and a group.

2nd group elements : Be (0.89), Mg (1.36), Ca (1.74), Sr (1.91) Ba(1.98)

17th group elements : F (0.72), Cl (0.99), Br (1.14), I (1.33)

3rd Period elements : Na(1.57), Mg(1.36), Al (1.25), Si(1.17), P(1.10), S(1.04), Cl(0.99)

4th period elements: K(2.03), Ca(1.74), Sc(1.44), Ti(1.32), V(1.22), Cr(1.17), Mn(1.17), Fe(1.17), Co(1.16), Ni(1.15), Cu(1.17), Zn(1.25), Ga(1.25), Ge(1.22), As(1.21), Se(1.14), Br(1.14)

3.5.2 Ionic radius

It is defined as the distance from the centre of the nucleus of the ion up to which it exerts its influence on the electron cloud of the ion. Ionic radius of uni-univalent crystal can be calculated using Pauling's method from the inter ionic distance between the nuclei of the cation and anion. Pauling assumed that ions present in a crystal lattice are perfect spheres, and they are in contact with each other therefore,

$$d = r_{C^+} + r_{A^-} \quad \text{----------} \quad (1)$$

Where d is the distance between the centre of the nucleus of cation C^+ and anion A^- and r_{C^+}, r_{A^-} are the radius of the cation and anion respectively.

Pauling also assumed that the radius of the ion having noble gas electronic configuration (Na^+ and Cl^- having $1s^2$ $2s^2$, $2p^6$ configuration) is inversely proportional to the effective nuclear charge felt at the periphery of the ion.

$$\text{i.e. } r_{C^+} \alpha \frac{1}{(Z_{eff})_{C+}} \quad \text{----------} \quad (1)$$

and

$$r_{A^-} \alpha \frac{1}{(Z_{eff})_{A^-}} \quad \text{----------} \quad (3)$$

Where Z_{eff} is the effective nuclear charge and $Z_{eff} = Z - S$

Dividing the equation 1 by 3

$$\frac{r_{C^+}}{r_{A^-}} = \frac{(Z_{eff})_{A^-}}{(Z_{eff})_{C^+}} \text{----(4)}$$

On solving equation (1) and (4) the values of r_{C^+} and r_{A^-} can be obtained

Let us explain this method by calculating the ionic radii of Na^+ and F^- in NaF crystal whose interionic distance is equal to 231 pm .